INCH-POUND
MIL-PRF-22885/116
26 July 2022

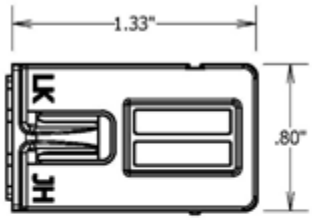
PERFORMANCE SPECIFICATION SHEET

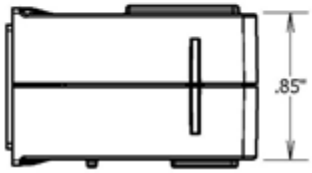
MODULE, OPTIONAL ELECTRONIC COMPONENTS (OEC) COMPATIBLE WITH MIL-PRF-22885/117, SWITCHING, LOGIC FUNCTION AND TERMINAL INTERCONNECTS, COMMON TERMINATION SYSTEM (CTS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the switches described herein shall consist of this specification and the latest issue of MIL-PRF-22885.

This specification covers the general requirements for Modules used as complementing functionality to the manually operated illuminated push button switches, switch assemblies as described on MIL-PRF-22885/108 and MIL-PRF-22885/108 and associated subassemblies as described on MIL-PRF-22885/117. Additionally, this specification cover a modules with the exclusive functionality of a Terminal Interconnect (Termial block).





NOTES:
1. Dimensions are in inches.
2. Unless otherwise specified, tolerances are ±.010 for three place decimals and ±.03 for two place decimals.
3. Each module requires a Common Termination System (CTS) connector that shall be designed and constructed to meet the performance requirements of MIL-PRF-22885/108 figure 10.
4. The CTS is M22885/10818440 per MIL-PRF-22885/108 and shall be acquired from a source listed on QPL-22885.
5. Exact shape of the module is optional provided dimensions specified are not exceeded.

FIGURE 1. Module A.

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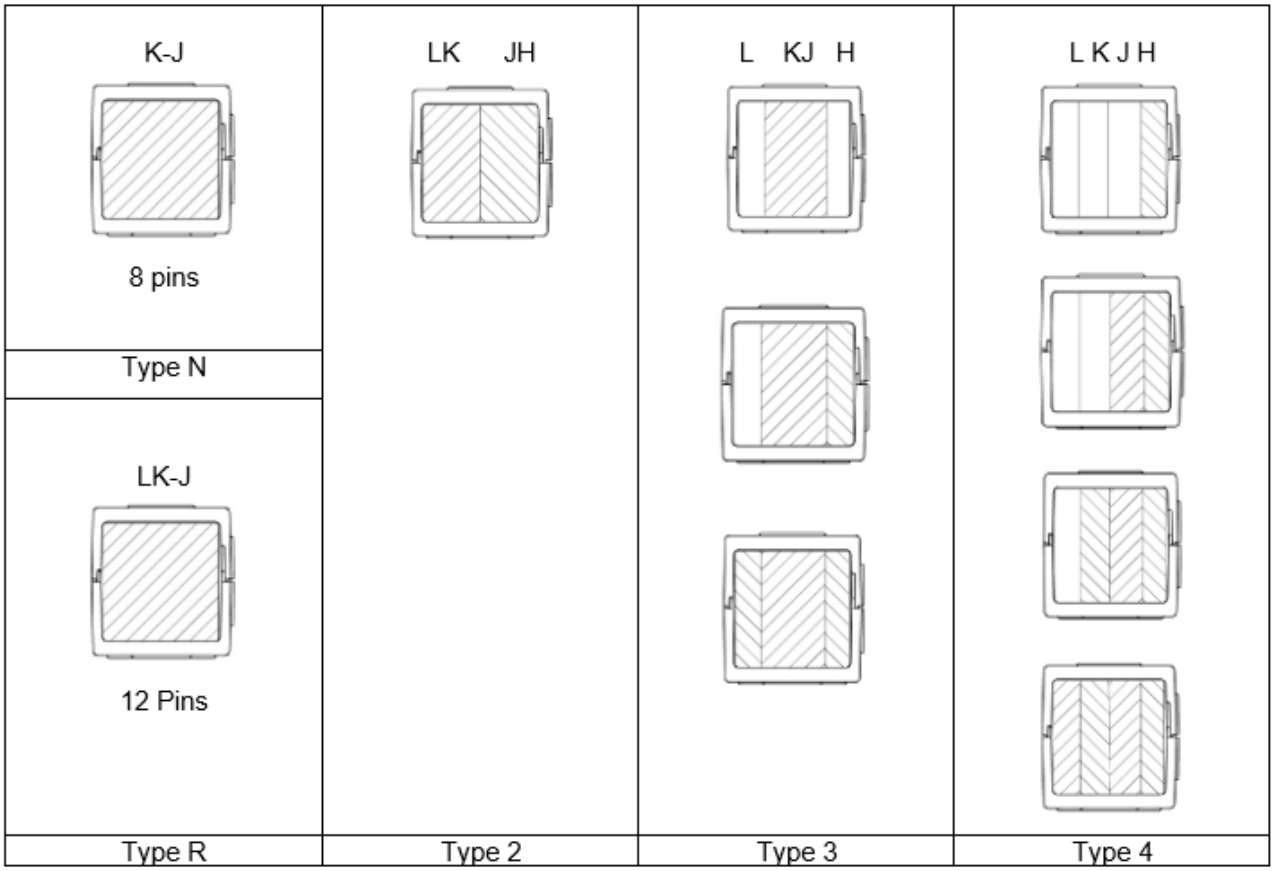


FIGURE 3. Types as defined on Table I.

TABLE I. OECs in the module.

Type	Size of Component allowed per type	Occupied Positions
N	1 Series N OEC required	Per Figure 3
R	1 Series R OEC required	
2	2 Series C OECs Required	
3	1 Series C OEC Required plus optional 0 to 2 Series A OECs	
4	1 Series A OEC Required plus optional 1 to 3 Series A OECs	

Notes:
1) See Table II for SERIES IDs
2) Refer to MIL-PRF-22885/117 for Technical definition of each Series of OEC

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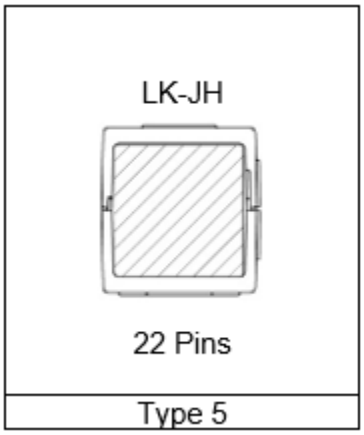


Figure 4. Type as defined in Table III.

TABLE II. Types of OECs in the module.

	Optional Electronic Component (OEC) Function	OEC Series	Mfg. ID	Mil Spec ID
Functional and Electrical Parameters per MIL-PRF-22885/117	Solid State Relay	A	SSR	4
	Combination- Solid State Relay 1/	C	SSRC	B
	Voltage Sensor 1/	A	VS	7
	Diode Pack	A	DP	3
	Terminal Block	A	TB	5
	Electronic Latch		EL	E
	Electronic Rotary	C	ER1	F
	Pulse Timer 1/	C	PT1	G
	Current Sensor 1/	A	CS	1
	Time Delay 1/	A	TD	6
	Square Wave Oscillator 1/	A	CT	2
	Defined Logic 1/	C	DL	D
	ARINC Single-Bit Converter 1/	N	SR429/1M	T
	ARINC Multi-Bit Converter 1/ 2/	N or R	SR429/4M	R
	ARINC Multi-Bit Binary Decoder 1/ 2/	N or R	SR429/4D	S

1/ These OEC have configurable options that will only be reflected on manufacturer part numbers.
2/ These OEC can be series N or R depending of the number of pins.

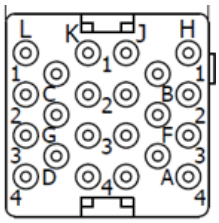
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Terminal Interconnects

Shall utilize the Module A from Figure 1 and the CTS as described in note 3 and 4 of Figure 1.

Termnal Interconnect splices the various positions of the Module up to max of 22 pins. See Table III.

TABLE III. Allowable Terminal Interconnects for Type 5 (see Figure 4).

 2/	Type	Mfg. ID	# Nodes 1/	Mil-spec ID
TB22: 22 (All Pins connected)	5	MPTB1	1	1
TB11:H1-4,B,F,A,J1-4 TB11: K1-4,C,G,D,L1-4	5	MPTB2	2	2
TB11: (H1-H4,B,F,A,J1-J4) TB7: (K1-K4,C,G,D) TB4: (L1-L4)	5	MPTB3	3	3
TB11: (H1-4,B,F,A,J1-4) TB4:(K1-K4) TB3: (C,G,D) TB4: (L-4)	5	MPTB4	4	4
TB4: (H1-4) TB3: (B,F,A) TB8: (J1-4,K1-4) TB3: (C,G,D) TB4: (L1-,L4)	5	MPTB5	5	5
TB4: (H1-4) TB3: (B,F,A) TB4: (J1-4) TB4: (K1-4) TB3: (C,G,D) TB4: (L1-,L4)	5	MPTB6	6	6

1/ Maximum Current rating is 7.5 Amps per node
2/ This drawing is for reference only and it is fully described on MIL-PRF-22885/108 figure 10.

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GENERAL REQUIREMENTS:

Design and Construction: See Figures 1 through 3

Functional Specifications: See MIL-PRF-22885/117

Materials:
Housing: High temperature thermoplastic.
Interconnect Pins: Per MIL-PRF-22885/117 and CTS per Figure 2.

Weight:
Module without CTS– 16 grams maximum
Module with CTS– 22 grams maximum
Module with CTS and Right-angle bracket – 32 grams maximum
Module with CTS and Right angle and Flush mount bracket – 42 grams maximum
Module with CTS and Right-angle bracket – 32 grams maximum

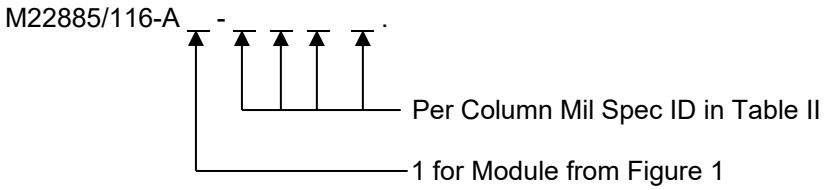
Installation Accessories:- Each module may include installation accessories such brackets or protective boots.

Shock II: High Impact Shock applicable only to OECs mounted in MIL-PRF-22885/108 pushbutton switches and MIL-PRF-22885/116 Logic Accessory Modules.

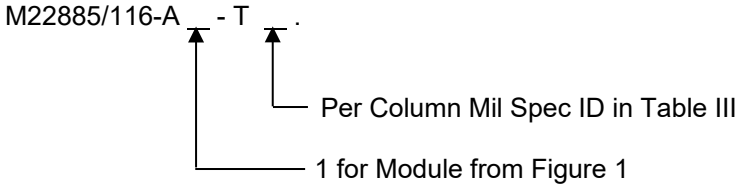
Electrical and EMC Requirements: See MIL-PRF -22885/117 for maximum levels per each OEC.

Part or Identifying Numbers (PIN): PIN's are assigned as follows:

Part Number for modules listed with components from Table II



Part Number for modules listed with Interconnects Table III



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TABLE IV. Group A Test Inspection.

Visual and mechanical examination (Size and weight) 1/
Operating Characteristics 2/
Dielectric Withstanding Voltage/3/4/5

1/ To be performed on each Lot and by Series of product.
2/ Functional test at typical operating voltage.
3/ Applicable between all leads and housing if metal surrounding OEC under test.
4/ Applicable between Terminal Interconnect Nodes.
5/ Dielectric withstanding voltage shall not be applied between OEC leads.

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TABLE V. Group B Inspection testing.

		Test Sample PIN's and Sample Numbers																		
		MFG PN 1		MFG PN 2		MFG PN 3		MFG PN 4		MFG PN 5		MFG PN 6		MFG PN 7				MFG PN 8		
		M22885/ 116 - xxxx		M22885/ 116 - xxxx		M22885/ 116 - xxxx		M22885/ 116 - xxxx		M22885/ 116 - xxxx		M22885/ 116 - xxxx		M22885/ 116 - xxxx				M22885/ 116 - xxxx		
Sample ==>		1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
Visual	per 4.7.1 of Mil PRF 22885	X	X	X	X	X	X	X	X	X	X	X	X	X	X		X	X	X	X
EMC/EMI per Table XXIX Mil-PRF-22885/117		X		X		X		X		X		X				X				
Shock I	MIL-STD-202-213		X								X			X				X		
Drying Period			X								X			X				X		
Insulation Resistance before Moisture	MIL-STD-202-302		X								X			X				X		
Moisture Resistance	MIL-STD-202-106		X								X			X				X		
Insulation Resistance after Moisture	MIL-STD-202-302		X								X			X				X		
Salt Spray	MIL-STD-202-101				X								X		X				X	
Electrical Endurance Inductive @ Altitude	per 4.7.28.2 of Mil PRF 22885 (when applicable)						X		X								X			X
Dielectric Withstanding @ Altitude	MIL-STD-202-301						X		X								X			X
Electrical Endurance Resistive @ Sea Level	per 4.7.28.2 of Mil PRF 22885 (when applicable)						X		X								X			X
Dielectric Withstanding Voltage	MIL-STD-202-301						X		X								X			X
Operating Charactersitics	per 4.7.6 of Mil PRF 22885	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
Dielectric Withstanding Voltage	MIL-STD-202-301		X								X			X				X		
Marking Visibility	MIL-STD-1285	X		X		X		X		X		X	X			X	X			X

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Referenced documents:
- MIL-PRF-22885
- MIL-PRF-22885/108
- MIL-PRF-22885/117
- MIL-STD-202-101
- MIL-STD-202-106
- MIL-STD-202-213
- MIL-STD-202-301
- MIL-STD-202-302
- MIL-STD-1285
- QPL 22885

Custodians:
Army - CR
Navy - EC
Air Force - 85
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Preparing Activity:
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(Project 5930-2022-043)

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